Percent Change Bar Model

Explore & Reflect

Name _____ Date _____ Period _____

Open **Percent Change Bar Model** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A $50 jacket is marked up 20%. Then the new price is discounted 20%. Is the final price $50?

2. What single number do you multiply by to increase a price by 25%?

2 Explore use the tool

1. Original $80, increase 25%. What is each 10% section worth, and what is the new amount?

2. Increase $80 by 20%, then decrease by 20%. Where do you end up, and why?

3. After a 25% increase, which second change brings the price exactly back to the start?

4. Find two increases in a row that make a total increase of exactly 32%.

3 Record use the tool and fill in the table

Percent changes

Use the second change when there are two.

Original	Change(s)	Multiply by	Final amount	Overall change
$60	−15%			
$40	+30%, then −10%			
$120	−25%, then +25%			
$200	+10%, then +10%			

4 Explain what did you discover?

1. A p% increase multiplies by _____. A p% decrease multiplies by _____.

2. In a second change, the percent is a percent of the _____ amount.

3. Two percent changes in a row _____; they do not add.

4. Explain why an increase of 50% followed by a decrease of 50% does not get you back to the original amount.

5 Apply a new situation, no tool

1. A $36 meal has a 15% tip added. What is the total?

2. A $250 bike is 30% off. Then 6% tax is added to the sale price. What is the final cost?

3. A price went from $40 to $46. What is the percent change?

4. After a 20% discount a shirt costs $32. What was the original price?